

1. (8%) What is the coefficient of $x^5y^3z^2$ in the expansion of $(x + 2y - z)^{10}$?

Ans: $C(10,5) \cdot C(5,3) \cdot C(2,2) \cdot 1^5 \cdot 2^3 \cdot (-1)^2$

2. (8%) You have a set of 100 distinct integers. What is the minimum number of integers you must select to guarantee that at least two integers have a difference that is divisible by 17?

Ans: 鴿籠原理

$17+1=18$

3. (8%) At a university with students from exactly 50 states, what is the smallest number of students needed to ensure that at least 100 students are from the same state?

Ans: 通用鴿籠原理

$\lceil x/50 \rceil = 100 \Rightarrow x = 4951$

4. (8%) How many distinct permutations are there of the letters in the word "BOOKKEEPER"?

Ans: $10!/(2!2!3!)$

5. (8%) Find the number of ways to make change for \$100 using \$10, \$20 and \$50 bills.

Ans: 10 (畫樹狀圖)

6. (12%) Suppose x_1, x_2, x_3 and x_4 are **nonnegative integers**.

(a) Find the number of solutions to $x_1 + x_2 + x_3 + x_4 = 7$

(b) Find the number of solutions to $x_1 + x_2 + x_3 + x_4 \leq 7$

Ans:

(a) $C(10,3)$

(b) $C(11,4)$

7. (12%) Suppose $|A| = 5$ and $|B| = 6$

(a) Find the number of functions $f: A \rightarrow B$.

(b) Find the number of **one-to-one** functions $g: A \rightarrow B$.

Ans:

(c) 6^5

(d) $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$

8. (12%)

(a) Find the **next 3** (緊接著3個) permutations in lexicographic order after 142563.

(b) Find the **next 3** 4-combinations of the set $\{1, 2, 3, 4, 5, 6, 7\}$ after $\{1, 2, 5, 6\}$.

Ans:

(a) 142635, 142653, 143256

(b) $\{1, 2, 5, 7\}, \{1, 2, 6, 7\}, \{1, 3, 4, 5\}$

9. (8%) In how many ways can we distribute 8 red balls and 10 blue balls among 5 children so that **each child receives at least one blue ball and 1 red ball**?

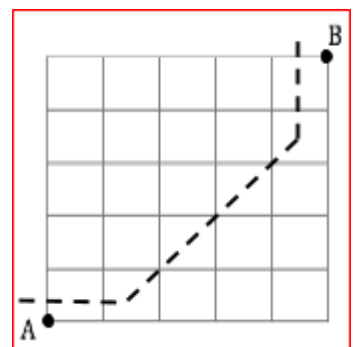
Ans: $C(7,4) \cdot C(9,4)$

10. (12%) The figure at the right shows a 5-block by 5-block grid of streets. Suppose that starting at the point labeled A you can go **one step up** or **one step to the right** in each move. This is continued until the point labeled B is reached.

(a) How many different paths go from A to B ?

(b) How many different paths go from A to B without crossing the dash line?

Ans: (a) $C(10,5) = 252$ (b) $C(8,3) - C(8,3) = 14$



11. (8%)How many ways are there to distribute 6 **different sandwiches** into 3 **identical lunch boxes** so that each box gets at least one?

Ans: 90

(stirling number)

$$S(n,j)=j \cdot S(n-1,j)+S(n-1,j-1)$$

j	1	2	3	4	5	6	7
n							
1	1						
2	1	1					
3	1	3	1				
4	1	7	6	1			
5	1	15	25	10	1		
6	1	31	90	65	15	1	

$$S(6,3)=3 \times S(5,3)+S(5,2)=3 \times 25+15=90$$